

Akhil Golepu

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PROFESSIONAL SUMMARY

Machine Learning MLOps Engineering student specializing in building end-to-end, production ready predictive systems. Proven ability to architect full-stack ML applications—integrating FastAPI, React, PostgreSQL—while ensuring model reproducibility through DVC and automated deployment pipelines. Driven by a professional interest in Satellite Image Processing and Geospatial Analytics, I aim to apply robust MLOps practices and advanced object detection frameworks to solve complex remote sensing challenges.

EDUCATION

Vardhaman College of Engineering

Bachelor of Technology in Computer Science and Engineering (AI & ML)

Hyderabad, India

Expected Jun 2027

– **CGPA:** 8.18 / 10.0

– **Relevant Coursework:** Artificial Intelligence, Machine Learning, Deep Learning, Computer Vision, Natural Language Processing, Data Structures & Algorithms, Object-Oriented Programming, Cloud Computing(AWS, Azure).

Ratna Junior College

Intermediate (MPC)

2023

– **Percentage:** 85.9%

Heritage Valley The Indian School

Class X (ICSE)

2021

– **Percentage:** 81.6%

TECHNICAL SKILLS

Machine Learning & DL: Scikit-learn, TensorFlow, PyTorch, Keras, CNNs, Transfer Learning, YOLO, Model Evaluation,

Data Science: Pandas, NumPy, Feature Engineering, Matplotlib, Seaborn

Computer Vision: OpenCV, Data Augmentation, Image Preprocessing, Image Segmentation, Feature Extraction

Programming: Python, SQL(PostgreSQL), Java

Tools & Platforms: DVC, MLFlow, Git, GitHub, Linux (Ubuntu), Google Colab, Jupyter Notebook

PROJECTS

AI-Powered ANPR System | *YOLO, EasyOCR* | [GitHub](#)

- Developed an end-to-end Automatic Number Plate Recognition system utilizing YOLOv8s (99.48% EasyOCR for recognition of Indian vehicle plates).
- Developed a domain-specific text normalization and post-processing layer to handle standardized Indian license plate formats, enabling automated cross-referencing and alerting against a centralized vehicle registry.

Customer Churn Prediction | *Python, CatBoost, FastAPI, React* | [GitHub](#)

- Implemented a persistent PostgreSQL schema with integrated Audit Logging to track prediction events, model changes, and user feedback for long-term system observability.
- Designed a Model Registry supporting 'Candidate', 'Shadow', and 'Active' model states, enabling structured model promotion and rollback capabilities.

Customer Segmentation | *Python, Scikit-learn, PCA, KMeans, GMM* | [GitHub](#)

- Built a behavioral segmentation engine using KMeans and GMM, providing cluster assignments with calibrated confidence score.
- Utilized PCA for dimensionality reduction and optimized clusters using the Davies-Bouldin index (1.02).

EXPERIENCE

Python Programmer Intern (MotionCut)

Sep 2024 – Oct 2024

Remote

Remote

- Developed custom Python modules for complex data processing and automation workflows.
- Optimized scripts and refactored codebase for improved maintainability and reliability.

CERTIFICATIONS

Machine Learning Specialization — Coursera

Artificial Intelligence and Machine Learning using Python — NIELIT Virtual Academy

Python Essentials 1 & 2 — Cisco